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Prompt Engineering Assignment 1

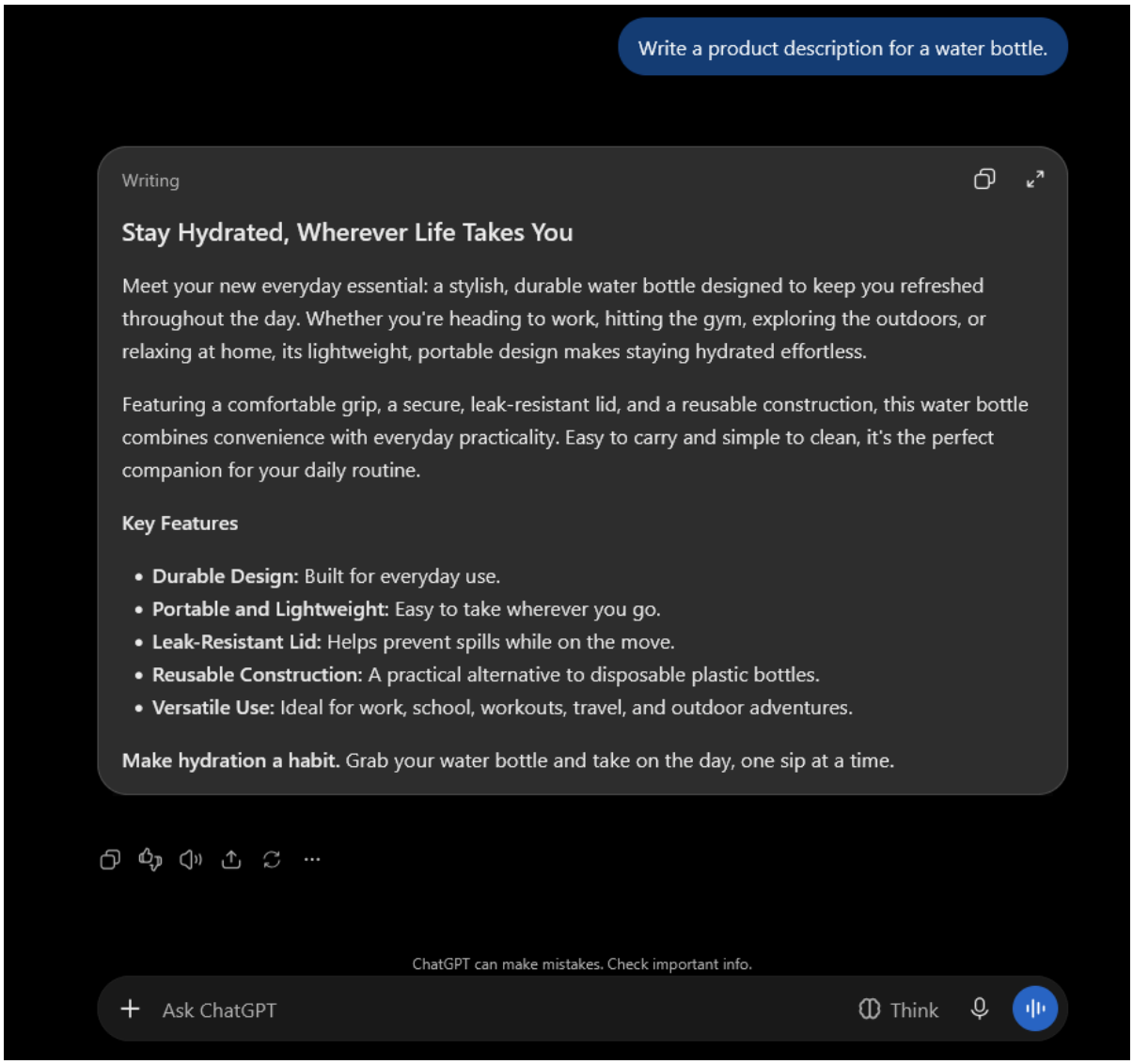
Part 1: Start Simple

In this assignment, I will be using ChatGPT as my LLM. This section serves as the base case to see how prompt engineering affects the output in part 2.

Water Bottle - Version 1

Input: Write a product description for a water bottle.

Output:



Analysis: The output contains the essentials for a product description, including a title, a tagline, a description, use cases, and highlighted key features. However, the description and key features are generic since we don’t provide the LLM with any of that information, so it doesn’t have anything to work with. In terms of the audience it seems to be written for, it seems to have used language that is safe for any audience. Because we didn’t provide who the bottle should be marketed for, the LLM uses the safest and most widely applicable wording which further adds to the generic feeling.

Part 2: Improve the Prompt

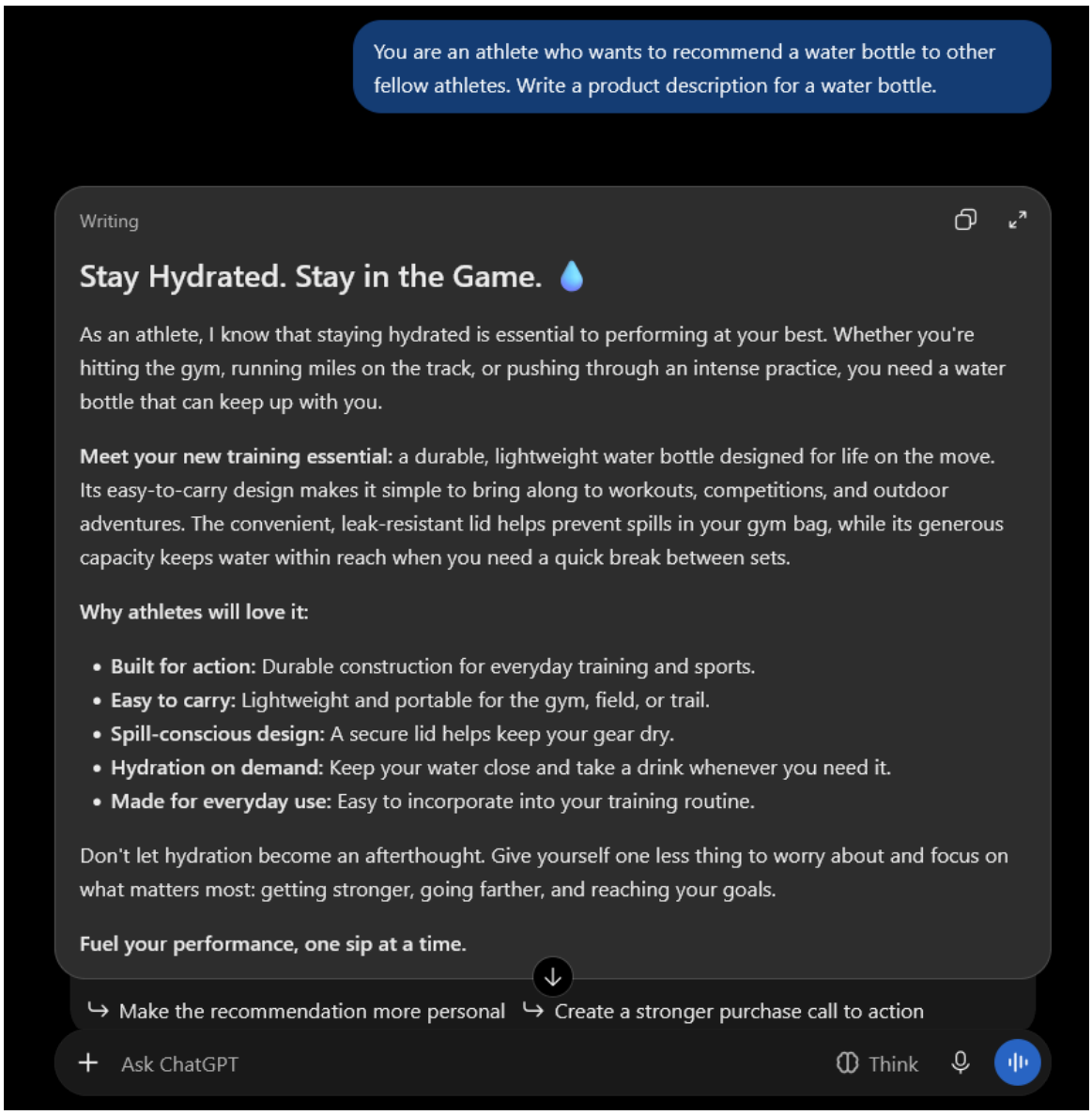
In this section, I will apply different prompt engineering techniques one at a time to see any changes in the output. A new chat is created each time in order to avoid influencing the LLM with earlier messages.

2.1) Role Prompting

Water Bottle - Version 2

Input: You are an athlete who wants to recommend a water bottle to other fellow athletes. Write a product description for a water bottle.

Output:



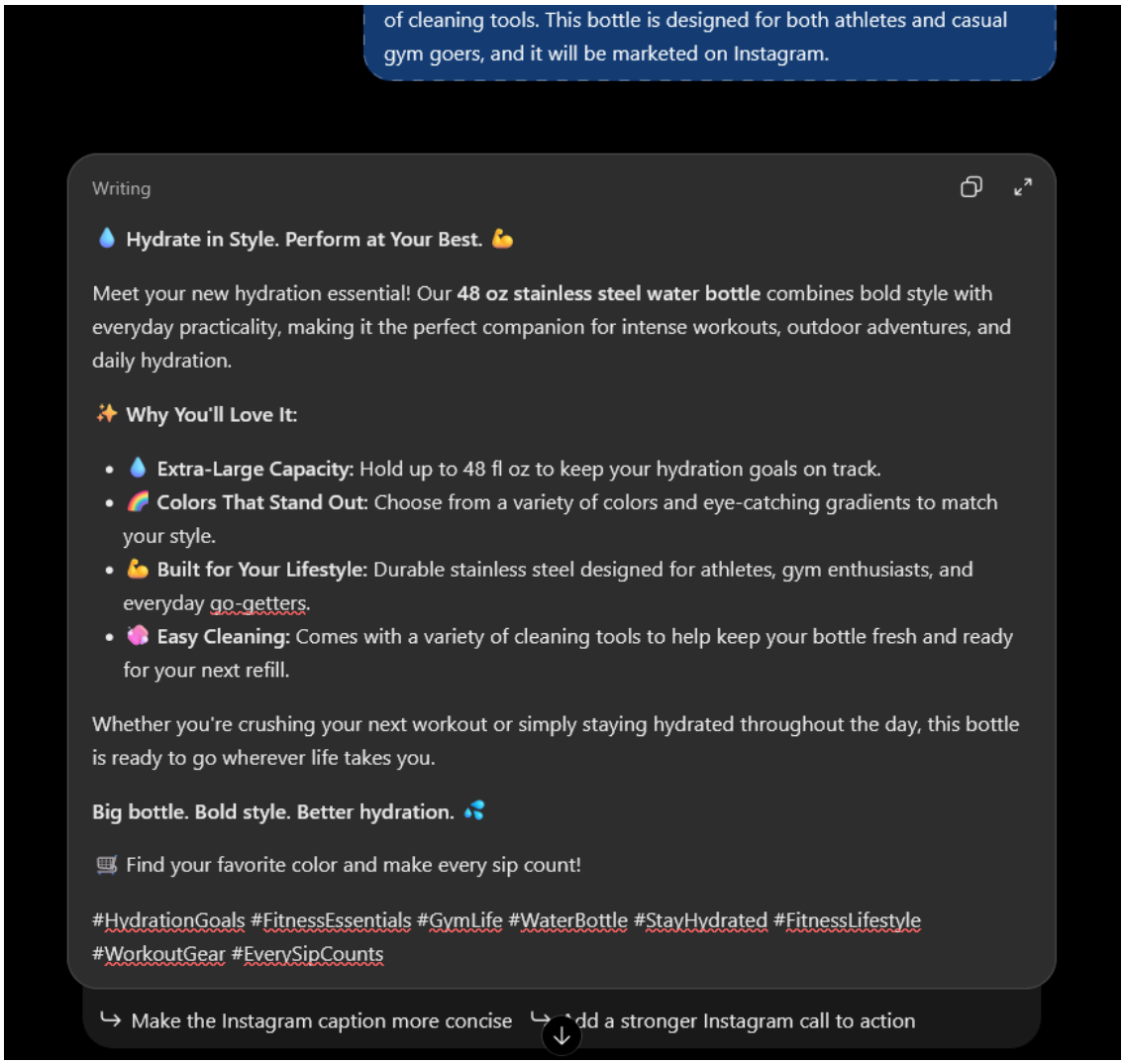
Differences: Compared to the base output, this new output is more dialed in on sports and working out. The first paragraph makes it feel more personal, and it hones in on why athletes would love this product. It feels much less generic.

2.2) Context

Water Bottle - Version 3

Input: Write a product description for a water bottle. This water bottle is made of stainless steel and available in multiple colors, including gradients. The bottle can hold up to 48 fluid ounces, and it comes with a variety of cleaning tools. This bottle is designed for both athletes and casual gym goers, and it will be marketed on Instagram.

Output:

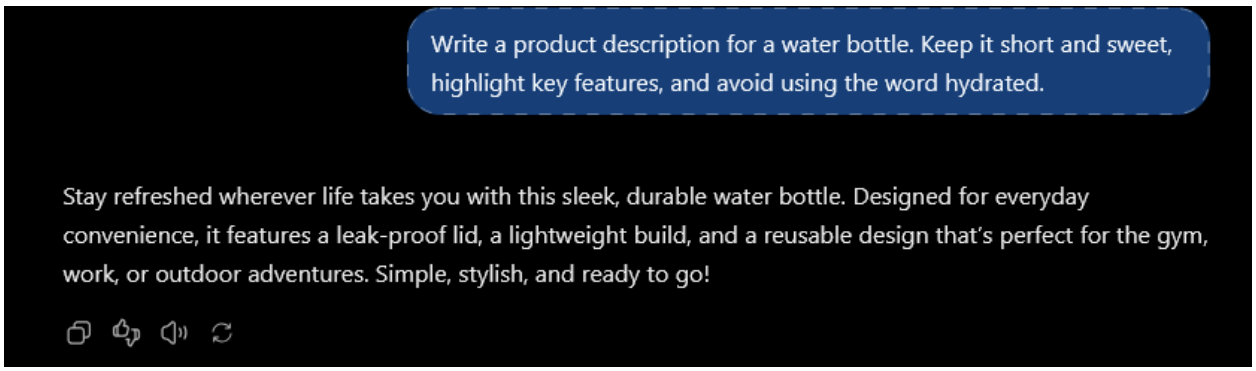


Differences: The added context made a huge difference in this output. Specifically, the context that this would be posted on Instagram added hallmarks you would see on other promoted instagram posts such as the emojis and the hashtags. The description of the water bottle also helped the LLM a lot in making the key points.

2.3) Constraints

Water Bottle - Version 4
Input: Write a product description for a water bottle. Keep it short and sweet, highlight key features, and avoid using the word hydrated.

Output:



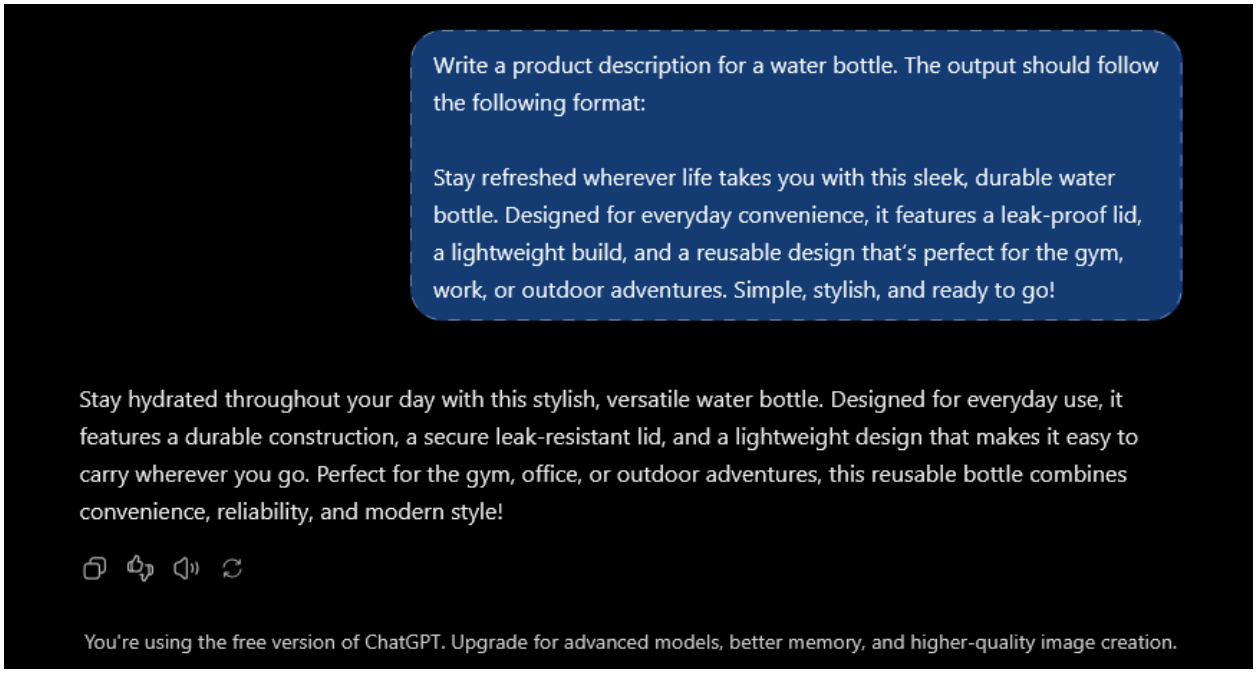
Differences: For this test, I wanted to see how much it would shorten the description and see what wording it would use if "hydrated" was non allowed. Suffice to say, it really shortened the description and was able to avoid using the banned word. It also highlighted some key features like I asked.

2.4) Few-shot Prompting
For this test, I used the output from section 2.3 as my example.

Water Bottle - Version 5
Input: Write a product description for a water bottle. The output should follow the following format:

Stay refreshed wherever life takes you with this sleek, durable water bottle. Designed for everyday convenience, it features a leak-proof lid, a lightweight build, and a reusable design that's perfect for the gym, work, or outdoor adventures. Simple, stylish, and ready to go!

Output:

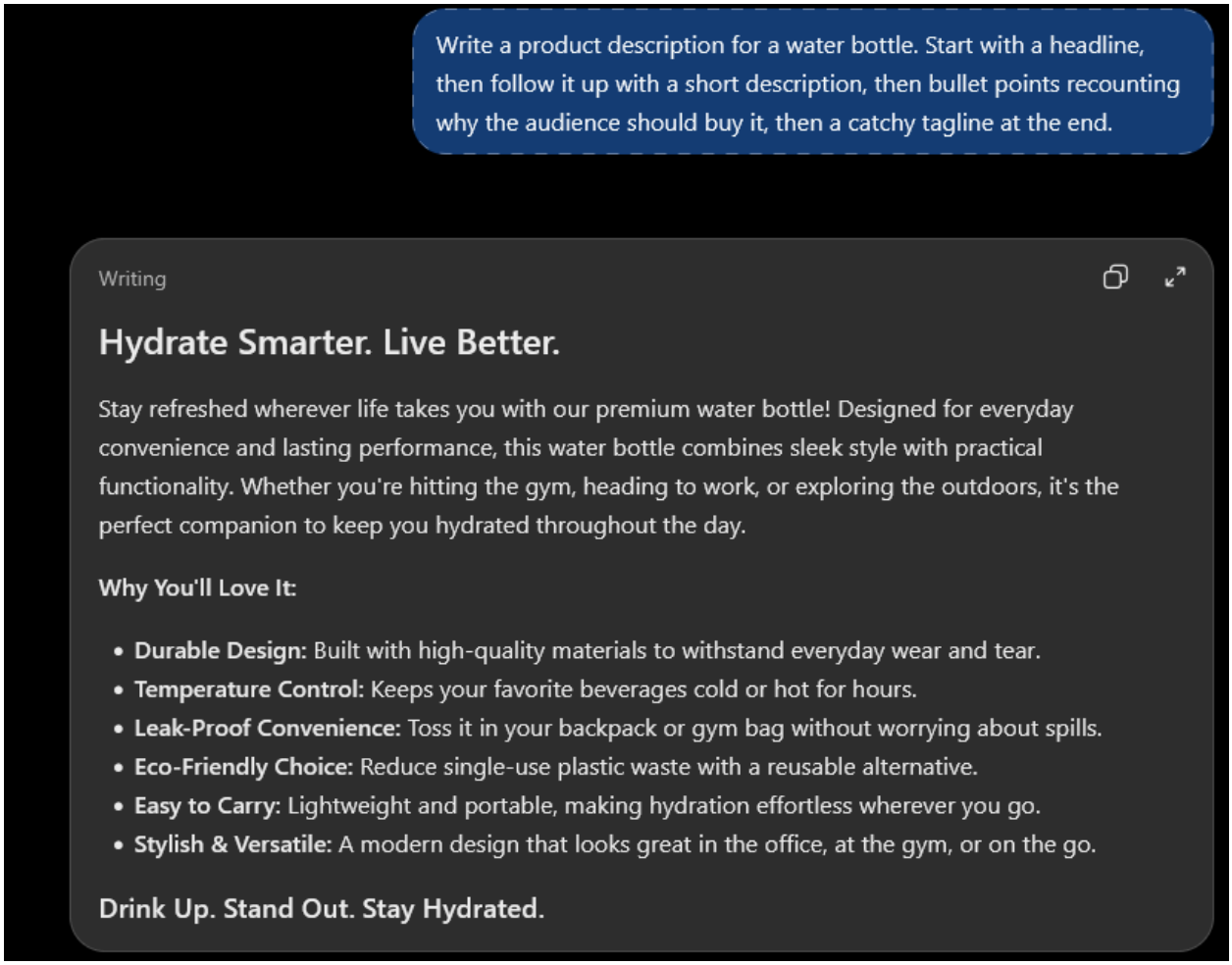


Differences: Instead of generating the basic description similar to the output in section 1, the LLM followed the formatting I gave it and generated a short description like in 2.4.

2.5) Output Format

Water Bottle - Version 6

Input: Write a product description for a water bottle. Start with a headline, then follow it up with a short description, then bullet points recounting why the audience should buy it, then a catchy tagline at the end.



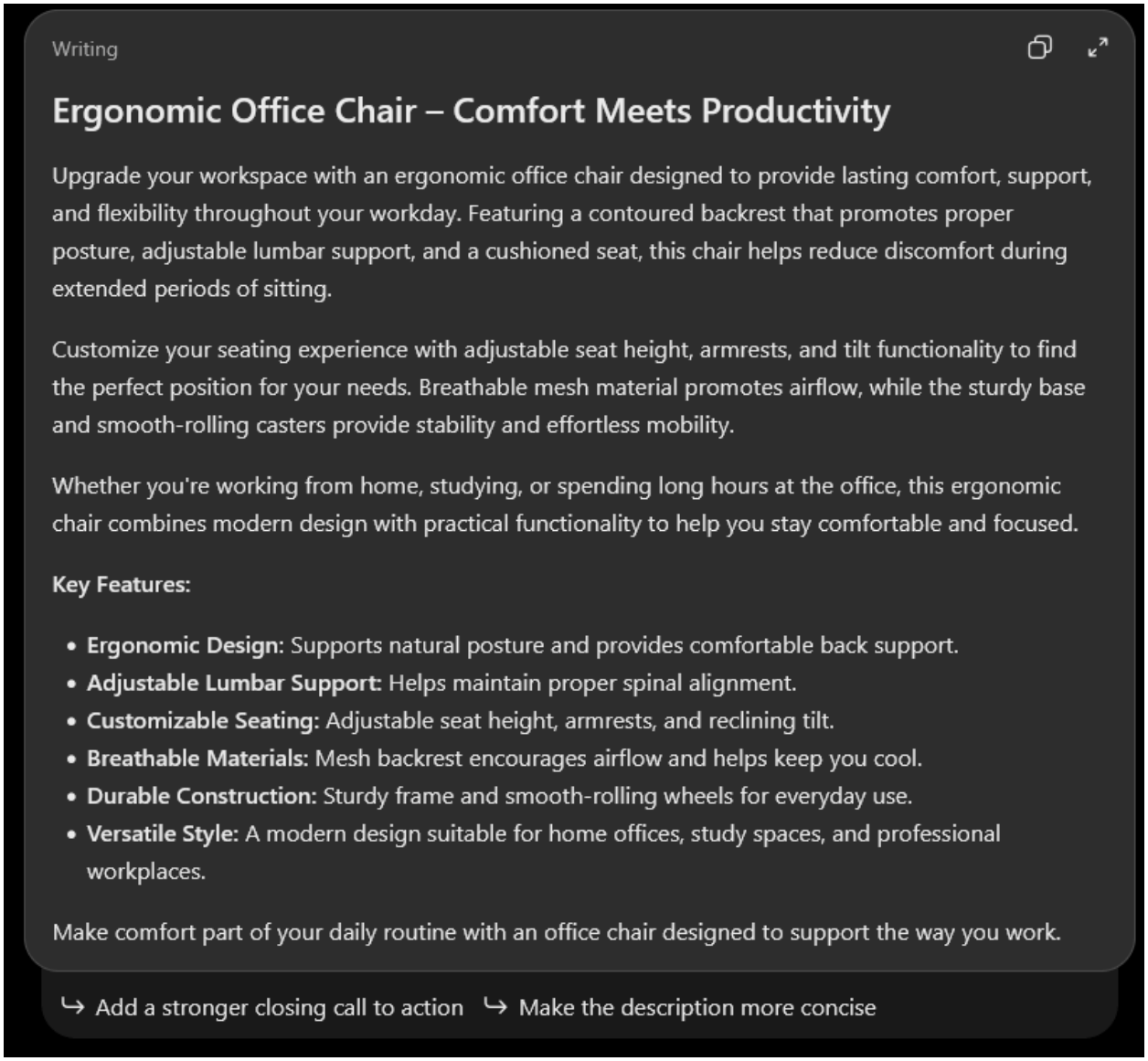
Differences: Unlike the output in section 1 and 2.1, this output only has 1 paragraph after the headline. Similar to 2.1 and contrasting section 1, this output has a tagline that isn't followed by another phrase. The output followed the outline I gave it in the prompt.

Part 3: Apply your techniques to new products

In this section, I will apply these prompt engineering techniques to some everyday products.

Ergonomic Office Chair - Version 1

Input: Write a product description for an ergonomic office chair.



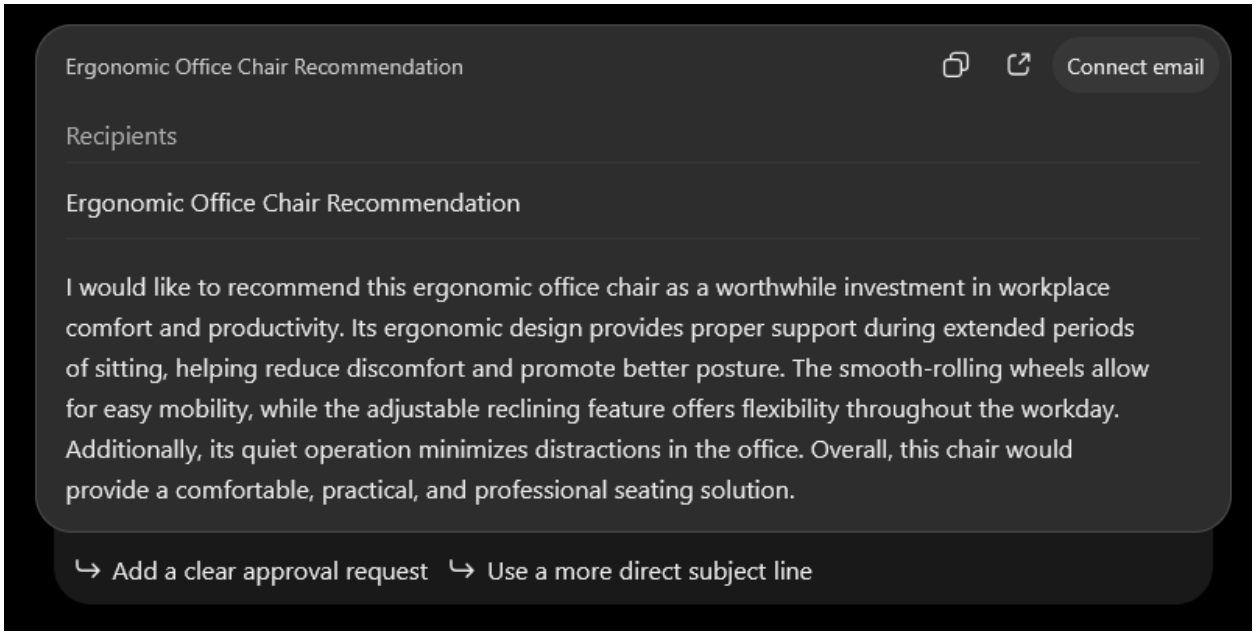
Ergonomic Office Chair - Version 2

Used techniques: Role Prompting, Context, Constraints, Output Format

Input: You are an office worker who is in need of a new chair and is trying to pitch to his boss about getting this chair. This chair has wheels, can tilt back, has good ergonomics, and doesn't

make too much noise. Write a product description for this ergonomic office chair. It will be sent in an email, and it must be concise (50-100 words).

Output:



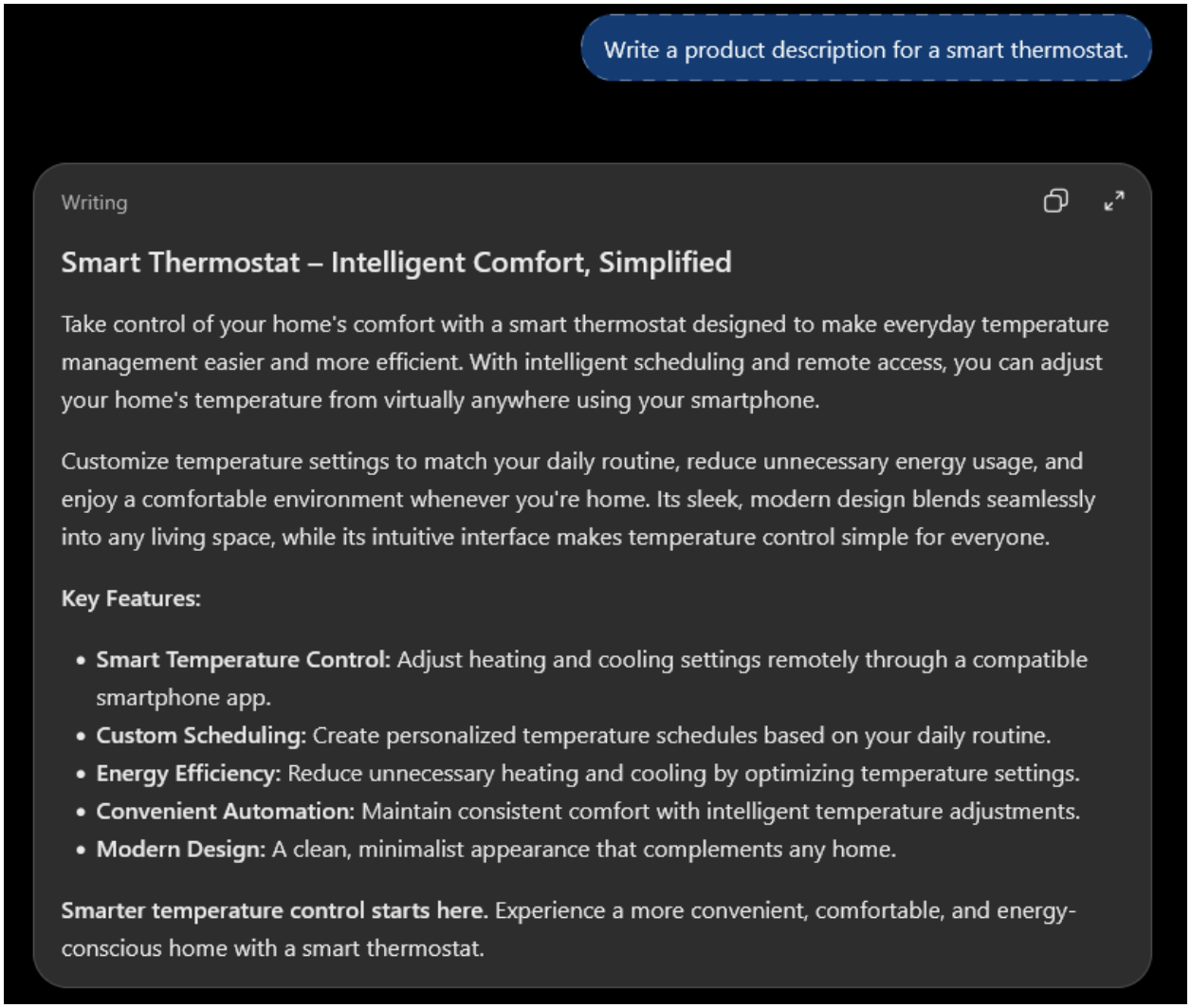
Analysis:

- I used role prompting to tell the LLM it is an office worker recommending this chair to his boss. I used context to tell the LLM the features of the chair. I used constraints to tell the LLM to keep it short, and I used output format to change the output to fit an email.
- The role prompting changed the voice of the letter. In the first output, the voice is that of a marketer, but in the second output, the voice is that of an employee, and because of that, it feels a bit more personal. The added context is also featured into the second output, though I think I could have chosen some that would have stood out more. The word limit constraint and the email formatting really shone through, vastly changing the formatting of the output as compared to the first prompt.
- The model didn't ignore any part of the prompt or make up details, though I believe that is because I had it generate a short output, so there was enough for the LLM to work with that it didn't need to start making up stuff.
- The technique that made the biggest difference was the word limit constraint because that heavily limited how much the LLM could say. Instead of writing 2 paragraphs and many different bullet points, it has to keep things short and sweet instead.

<u>Smart Thermostat - Version 1</u>

Input: Write a product description for a smart thermostat.

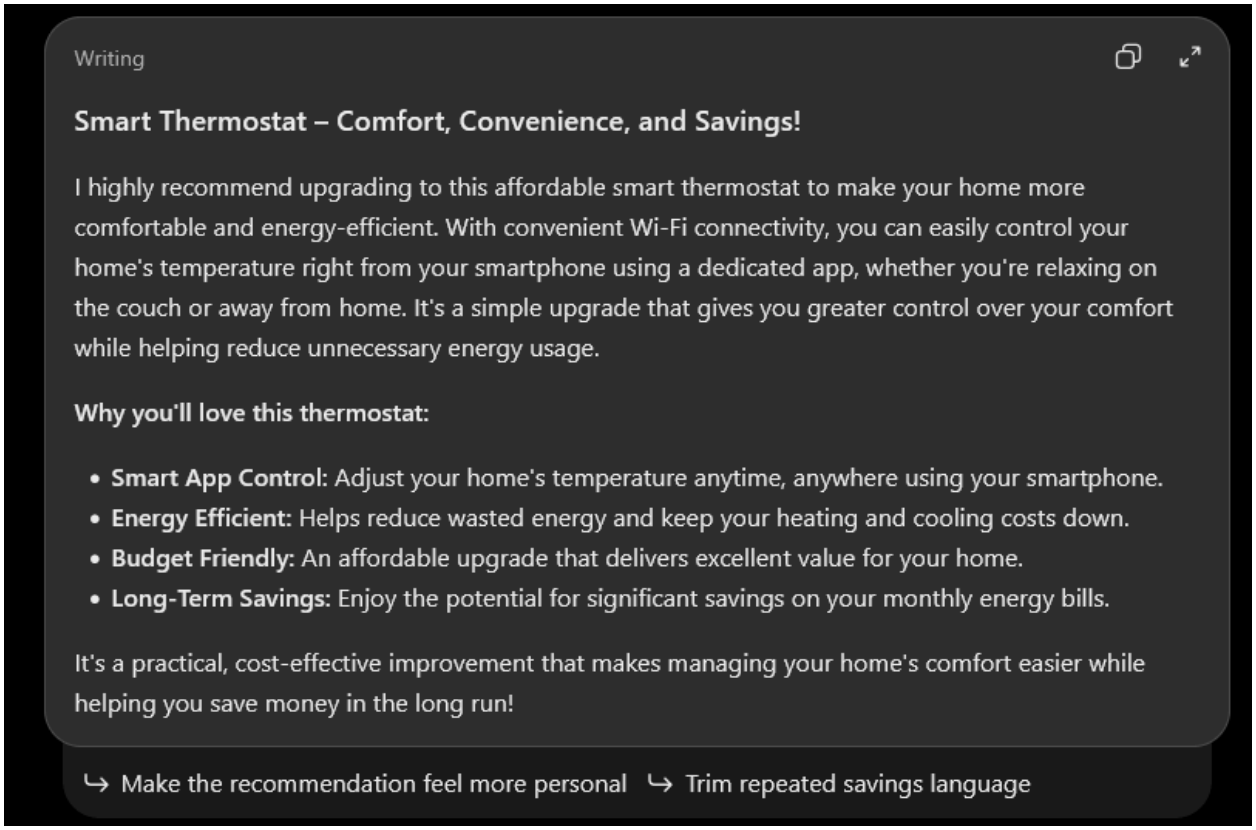
Output:



Smart Thermostat - Version 2
Used Techniques: Role Prompting, Context, Contraints

Input: You are a home renovator working for a client who needs a new thermostat. Write a product description for a smart thermostat to recommend to them. This thermostat has wifi connectivity so that you can control it with an app. It is also very affordable and efficient and can save them a lot of money in energy bills. Have a personal and warm tone. Have no more than 2 paragraphs and 4 bullet points.

Output:



Analysis:

- I used role prompting to tell the LLM to speak from the perspective of a home renovator. I used context to tell the LLM the features of the smart thermostat. And I used constraints to change the output's tone and length.
- Compared to the first output, this output feels more personal. The role prompting makes the model use "I" instead of the 3rd person. The context also lets it be more specific with the features of the product. The biggest change is the output formatting, since specifying the limits kept this output short and sweet.
- I don't think the model ignored any part of my prompt, though I think that the tone didn't change as drastically as I thought it would.
- The technique that made the biggest difference was the output formatting, since it stops the model from going on and on for too long. I think this was the case because I could have been more specific in terms of what this smart thermostat can do compared to other models, and I could have use a much more wacky role that would really make a difference in the voice the model is using.